One Free Number: An Exhaustive Closure on the Normalization of the de Sitter–MOND Acceleration Scale $a_0 = c^2\sqrt{(\Lambda/32\pi)}$

Carl P. Zimmerman · Briar Creek Tech · 2026-06-26

Abstract

The MOND acceleration scale $a_0 \approx 1.2\times10^{-10}$ m s^{-2} coincides numerically with the cosmological quantity $c\sqrt{\Lambda}$, and can be written exactly as a horizon surface gravity,

$$\mathbf{a_0 = c^2\sqrt{(\Lambda/32\pi)} = (c/2)\sqrt{(G\rho\_DE)} = cH\_\Lambda \,/\, Z,\ Z = \sqrt{(32\pi/3)} \approx 5.789.}$$

The dimensionless content of this identity factorizes cleanly. The group **$\sqrt{(8\pi/3)}$ is forced** — it is the Einstein coupling 8π (from $\rho\_DE = \Lambda c^2/8\pi G$) times the Friedmann 3 (from $H^2 = 8\pi G\rho/3$), a sympy-exact algebraic fact. The **only** undetermined number is the outside coefficient **κ = ½** (the "c/2"). Whether a_0's *value* is derived or merely fitted therefore reduces to a single yes/no question: can κ = ½ be forced from first principles?

This note reports an exhaustive negative answer. Across **~24 structurally-distinct routes** — ghost-freedom, unitarity, holography, a microscopic degree-of-freedom count, the Atiyah–Patodi–Singer η-invariant on the de Sitter horizon, five "symmetry-breaking" mechanisms (Hopf-bundle flux, the Witten SU(2) anomaly, the gravitational-Chern–Simons framing anomaly, a cosmic-string conical defect, a ζ-regularized Casimir energy), eight further families (on-shell dS-Unruh, 4-form flux quantization, horizon-entropy quantization, and five further normalization, index, and limit-matching constructions), and a gated brute-force over six classes of modified-inertia action — **no construction forces κ = ½ non-circularly.** Every route fails for one structural reason, which we make precise: κ is the absolute classical normalization sitting *outside* the gravitational square-root, while every number-fixing probe in physics (signs, ratios, scales, quantized levels, spectral asymmetries, vacuum energies) lives *inside* it, or is a mod-$\mathbb{Z}$ / dimensionless object **type-mismatched** to a bare multiplicative coupling. The mismatch is formal: $a_0(2\kappa) = 2\cdot a_0(\kappa)$ exactly, so κ is strictly linear and classical, while every quantized invariant is periodic or rational. The one type-distinct avenue — asymptotic safety, which outputs real numbers rather than mod-$\mathbb{Z}$ invariants — is examined separately and also fails: a_0 depends on $\sqrt{\Lambda}$ alone, so the renormalization-group fixed point's G–Λ invariant is orthogonal to it.

A genuine and instructive caveat closes the loop. The framework *does* contain a rigorously forced ½ — the binomial ½ in the de Sitter–Unruh effective acceleration $\sqrt{(a^2+(cH)^2)} - cH = a^2/(2cH) + \ldots$ — but it is bonded to cH and therefore sets the **temperature root** $a_0 = 2cH\_\Lambda$. The two roots differ by **exactly 2Z ≈ 11.58** (the long-noted "~12×"); forcing this ½ into κ's slot would demand $\kappa\_eff = Z \approx 5.789$, not ½. It is a real ½ of the framework's own making, and it is provably not κ.

The defensible conclusion is unchanged from, and strengthened relative to, the author's prior work: **$a_0 \approx c\sqrt{\Lambda}$ is a forced scale, while its precise normalization is one free posit. The**

framework is a provably one-parameter effective theory — not zero-parameter, and not a theory of everything. What we add here is not a new mechanism but the demonstration that, within the established toolkit, there is a structural reason no such mechanism appears. The live content is now entirely empirical: a declining $a_0(z) \propto \sqrt{\rho\_DE(z)}$ and a Lorentz-violating s^TX dipole, both falsifiable this decade. All symbolic checks are reproducible from the public repository.

---

1. What is forced, and what is one number

Write the de Sitter rate $H\_\Lambda = c\sqrt{(\Lambda/3)}$ and the dark-energy density $\rho\_DE = \Lambda c^2/8\pi G$. The acceleration scale is

$$a_0{=}\kappa\, c\sqrt{G\rho_{\mathrm{DE}}}{=}c^2\sqrt{\frac{\Lambda}{32\pi}} \quad \text{at} \;\; \kappa = \frac{1}{2}, \qquad \rho_{\mathrm{DE}} = \frac{\Lambda c^2}{8\pi G}.$$

Equivalently $a_0 = cH\_\Lambda / Z$ with $\mathbf{Z = cH\_\Lambda / a_0 = 2\sqrt{(8\pi/3)} \approx 5.789}$. Because $H^2 = 8\pi G\rho/3$ holds for *any* density, the ratio $Z = \sqrt{(32\pi/3)}$ is an algebraic identity, not a measurement: the H cancels. Its dimensionless value decomposes as

$$Z = 2\sqrt{\frac{8\pi}{3}}, \qquad 8\pi \text{ (Einstein: } \;\rho_{\mathrm{DE}} = \Lambda c^2/8\pi G)\times\frac{1}{3} \text{ (Friedmann: } \;H^2 = 8\pi G\rho/3),$$

with **the lone free factor the outside $2 = 1/\kappa$.** Everything physical about whether a_0's number is *derived* lives in this single coefficient. The question of this note is therefore sharp and binary.

2. The exhaustive closure

We organize ~24 attempts to force $\kappa = ½$ into six structural failure-classes. Each was checked by direct symbolic computation; the per-route verdicts and scripts are in the repository.

Class	Representative routes	Why it cannot reach κ
A. Scale-fraction wall	ghost-freedom; unitarity; holography; topological Dirac-η; the dS-Unruh Taylor-½	sees only signs / ratios / scales / a mod-$\mathbb{Z}$ phase — never the outside multiplier
B. Double-count	de Sitter surface gravity $c^2/2R$; Gibbons–Hawking T, S = A/4; ζ-Casimir; 4-form flux energy	the ½ is real but *already spent* building cH_Λ and the 8π in ρ_DE
C. κ cancels from the ratio	4-form quantization integer n; CKN g* dof-count; the deep-Newtonian g_obs/g_bar ratio	κ divides out of every observable ratio ⟶ unconstrained
D. Inserted structure	Hopf base-flux + parity anomaly; Witten SU(2); a posited cosmic string	the ½ needs a field/flux de Sitter does not force (MM curvature F = 0 on-shell)
E. Tunable, not ½	lens-space η; cosmic-string deficit angle; CKN root g*	gives a continuum fixed by a free scale, ½ only at a hand-picked value
F. Conventional / mod-$\mathbb{Z}$ phase	round-S^3 Dirac η = 0; Chern–Simons level ∈ $\mathbb{Z}$; framing c/24; θ/2π; APS η/2	a periodic or geometry-blind object, not a classical coupling

Two computations deserve a sentence. **The de Sitter horizon is the Hopf bundle** $S^1 \to S^3 \to S^2$; one might hope a base-S^2 monopole flux ($H^2(S^2)=\mathbb{Z}$, unlike $H^2(S^3)=0$) supplies a ½. But the de Sitter saddle is the *full* S^3 (the round Dirac η is exactly 0 by the ± symmetric Camporesi–Higuchi spectrum), the MacDowell–Mansouri SO(4,1) curvature **vanishes on-shell** (de Sitter is the flat connection, so there is no dynamical flux to twist), and the only ½ available — a Chern–Simons parity-anomaly level — is mod-$\mathbb{Z}$ and requires an inserted charged fermion. The forced Hopf numbers are 0 and 1, never ½. Separately, **the Witten SU(2) anomaly** (often invoked via “3 generations is odd”) in fact counts electroweak *doublets*: the Standard Model has (3 colors + 1 lepton)×3 = 12, an *even* number, so the anomaly never fires.

3. The structural reason

The failures are not independent accidents; they are one fact seen six ways. Any candidate dynamical operator splits into a **functional shape**, which lives *inside* the gravitational root and *can* fix a clean number — the conformal coupling $\xi = 1/6$, the a_4 heat-kernel anomaly, the binomial ½, and Z itself are all genuinely forced — and an **overall normalization**, which lives *outside* the root, and is where κ resides. The separation is exact:

$$\frac{a_0(2\kappa)}{a_0(\kappa)} = 2.$$

κ is a strictly **linear, classical, *ħ*-free multiplier.** Every number-fixing tool physics offers is, by contrast, a mod-$\mathbb{Z}$ level (Chern–Simons, framing, θ), a spectral asymmetry mod $\mathbb{Z}$ (η-invariants), a dimensionless ratio (unitarity, ghost-freedom), or a vacuum energy that renormalizes ρ_DE (Casimir, flux). None of these can equal a bare multiplicative coupling — a periodic or rational object cannot fix a linear one. Physically, the value $a_0 = c^2\sqrt{(\Lambda/32\pi)}$ depends on √Λ *alone*: the Newton constant cancels identically, since ρ_DE = $\Lambda c^2/8\pi G$ makes √(Gρ_DE) = c√(Λ/8π). The Einstein factor 8π survives *inside* the root as part of the forced shape; κ is the bare prefactor *outside* it, carrying no coupling at all. This is why κ is doubly out of reach — no spectral, topological, or ratio probe can fix a linear classical multiplier (the type-mismatch above), and no ultraviolet-gravity completion acting *through* G can reach a quantity from which G has already cancelled. The brute-force over six independent action classes (k-essence, nonlocal kernels, modified dispersion, entropic, Horndeski, spectral) returns no survivor for the same reason: each leaves κ a free outside multiplier, selects the temperature root, or inserts the answer.

4. The forced ½ that exists — and is not κ

It would be dishonest to report only the negative. The framework contains a **genuinely forced ½**, and locating it sharpens the result. The de Sitter–Unruh modified-inertia mechanism (Milgrom 1999; Deser–Levin 1997) gives the effective low-acceleration response

$$g_{\text{eff}} = \sqrt{a^2+(cH)^2} - cH = \frac{a^2}{2cH} - \frac{a^4}{8c^3H^3} + \cdots,$$

whose leading coefficient is **exactly 1/(2cH)** — the universal $\sqrt{(1+x^2)}-1 \approx x^2/2$ half, forced by the geometry, not inserted. But this ½ multiplies 1/cH: it is bonded to the de Sitter *rate* (the Friedmann/temperature root) and yields a_0 = 2cH_Λ. The coefficient κ we need multiplies the *density* root √(Gρ_DE), which carries the Einstein 8π. The two roots differ by exactly Z:

$$\frac{a_0^{\text{temp}}}{a_0^{\text{dens}}} = \frac{2cH_\Lambda}{cH_\Lambda/Z} = 2Z = \frac{8\sqrt{6}\,\sqrt{\pi}}{3} \approx 11.58.$$

So the forced ½ is real and is the framework's own — but it is the *temperature-root* ½, separated from κ by exactly the factor Z = 5.789. Forcing it into κ's slot would require κ_eff = Z ≈ 5.789, not ½: a Z-error, sympy-exact. The intuition that "a forced ½ is in there" is correct; it simply belongs to the other root.

5. Scope and consequence

The result is a **strong negative**: no non-circular construction forces κ = ½ across ~24 structurally-distinct routes, and the reason is a precise type-mismatch rather than a failure of effort. It is *not* a formal no-go theorem against every conceivable exotic boundary condition or operator; an exhaustively-excluded statement of that strength is not claimed. What is claimed, and is doubly robust, is that within the established toolkit the ½ either is not present on the de Sitter geometry, or — even where a ½ appears — is not the outside classical coefficient κ.

One avenue deserves explicit mention, because it is *not* of the mod-$\mathbb{Z}$ kind addressed above. Asymptotic safety — the conjecture that gravity reaches an interacting ultraviolet fixed point — outputs genuine real numbers (the fixed-point couplings g^*, λ^* and the critical exponents), so the type-mismatch does not block it, and it was examined on its own footing. It too fails to force κ. Decisively, a_0 depends on $\sqrt{\Lambda}$ alone — G cancels — whereas asymptotic safety's one scheme-robust invariant is the product $g^*\lambda^* \propto G\Lambda$, *orthogonal* to a_0's argument and numerically ≈ 0.10–0.14 across renormalization schemes (nearer $1/Z \approx 0.17$ than ½, and scheme-dependent where κ = ½ is exact); moreover its infrared limit is general relativity with a constant Λ, not a modified-inertia phase. We record asymptotic safety, explicitly, as the one theoretical avenue closed by *no non-circular construction found* rather than by a formal no-go.

The consequence is clean and was the author's published position, here proven rather than asserted: **the framework is a provably one-parameter effective theory.** Its forced content (the scale $a_0 \approx c\sqrt{\Lambda}$, the ratio $Z = 2\sqrt{(8\pi/3)}$, the de Sitter–Unruh square-root law, and with it the Radial Acceleration Relation and the Baryonic Tully–Fisher Relation) stands untouched. Its one free number, κ = ½, is now understood to be free *for a reason.* This is a stronger statement than "the mechanism was not found": there is a structural obstruction to there being one.

The live program is therefore entirely empirical. The single distinctive prediction — a declining acceleration scale, $a_0(z) \propto \sqrt{\rho\_DE(z)}$, with $a_0(z{=}3) \approx 0.74\, a_0(0)$ and a corresponding negative high-redshift Baryonic-Tully–Fisher offset — is falsifiable this decade with DESI w(z), JWST/ALMA high-z disc kinematics, and the ELT. A second, independent handle is the framework's forced Lorentz-violating signature: a gravitational s^TX boost-dipole at the few-$\times 10^{-9}$ level, within reach of Gaia DR4-era solar-system and astrometric bounds. The math is mapped; the verdict now belongs to the data.

Reproducibility. All symbolic and numerical results are reproducible from the public repository, `real_research/` verdict documents `TOPOLOGICAL_KAPPA_ETA_VERDICT`, `KAPPA_FIVE_HAIL_MARY_VERDICT`, `KAPPA_ALL_DOORS_VERDICT`, and `KAPPA_GATED_ACTION_BRUTEFORCE_VERDICT`. This note documents a closure and corrects nothing in the author's prior honest description; it claims *less*, not more — a one-parameter effective theory with a now-understood reason for its single free coefficient.

Key references. Milgrom (1983, 1999, 2011); Deser & Levin (1997); Atiyah, Patodi & Singer (1975); Witten (1982, 1989); Camporesi & Higuchi (1996); Gibbons & Hawking (1977). Supersedes-toward-less and supplements arXiv/Zenodo 10.5281/zenodo.20935948.